

A Joint Embedding Predictive World Model for JAXAtari

Implementing and evaluating LeWorldModel on the JAXAtari-15 suite

Amirmohammad Raei
Praktikum in der KI, Topic 30 (Deep model based RL: JEPA and Dreamer)
Technische Universität Darmstadt

September 2026

Abstract

This report describes an implementation of LeWorldModel (LeWM), a joint embedding predictive architecture for learning world models directly from pixels, and its evaluation on the fifteen games of the JAXAtari suite. No JAX or JAXAtari implementation of the method existed, so the model, the training loop, the evaluation protocol and the downstream agent were all written from scratch for this lab. The world model trains without representation collapse on all fifteen games and produces useful open loop latent predictions on fourteen of them. A set of seven single variable ablations tests the paper’s central architectural claim, that the SIGReg regulariser removes the need for a stop gradient. The claim is partly supported and partly narrowed: removing SIGReg does collapse the representation, but a plain stop gradient also prevents collapse, so SIGReg is sufficient rather than necessary, and what it actually contributes is a substantially better predictor. Finally, a PPO agent is trained on the frozen representation and compared against an identical agent trained from raw pixels. Averaged over three seeds on Pong, the pretrained representation roughly halves the number of environment steps needed to reach positive play and reduces run to run variance by a factor of about seven, while the final score after one million steps is not significantly different.

1 Introduction

Model based reinforcement learning agents learn a model of the environment and use it to plan or to shape a policy. Most such models predict future observations in pixel space, which forces the model to spend capacity on visual detail that is irrelevant to control. Joint embedding predictive architectures (JEPAs) take a different route. They predict the *embedding* of a future observation rather than the observation itself, so that anything the encoder chooses to discard is also removed from the prediction target.

That design has a well known failure mode. If the encoder is free to choose the target, it can drive the prediction loss to zero by mapping every input to the same constant vector. The representation then carries no information at all while the training loss looks excellent. Methods in this family therefore need some mechanism that prevents collapse. BYOL and its descendants use a stop gradient on the target branch together with an exponential moving average target encoder.

LeWorldModel (LeWM) proposes an alternative. It adds a distributional regulariser called SIGReg, a sketched isotropic Gaussian regulariser, and claims that this term alone is enough to keep the representation open, so that the whole system can be trained end to end with no stop gradient and no separate target network. That claim is architecturally interesting because it simplifies the training setup considerably, and it is the claim this report sets out to test on a new domain.

The task for this lab was to implement the algorithm on JAXAtari, make it run on the fifteen required games, and compare the results against existing implementations. Since the

method predicts embeddings and never predicts reward, it produces no game scores by itself, so an additional downstream evaluation was built in order to obtain numbers that can be compared with published reinforcement learning baselines.

2 Background

2.1 The objective

Let f_θ be an encoder mapping a single frame to an embedding, and let g_ϕ be a predictor that maps a sequence of embeddings and the actions taken to a prediction of the next embedding. Writing $z_t = f_\theta(x_t)$, the training objective is

$$\mathcal{L} = \underbrace{\frac{1}{T} \sum_t \|g_\phi(z_{1:t}, a_{1:t}) - z_{t+1}\|^2}_{\text{prediction}} + \lambda \cdot \underbrace{\text{SIGReg}(z)}_{\text{anti collapse}}. \quad (1)$$

The important detail is that the target z_{t+1} is produced by the *same* encoder, and gradients flow through it. There is no stop gradient and no target network. Collapse is prevented only by the second term.

2.2 SIGReg

SIGReg measures how far the distribution of embeddings is from an isotropic Gaussian. It draws a large number of random one dimensional projections of the embedding batch and applies an Epps and Pulley style normality test to each projection, comparing the empirical characteristic function of the projected data against that of a standard normal. The statistic is averaged over projections. Because an isotropic Gaussian occupies all directions of the embedding space equally, minimising this term directly opposes the concentration of embeddings into a low dimensional subspace, which is what collapse looks like geometrically.

The implementation of this term follows the reference code of the paper closely, since it is the component the paper’s contribution rests on and any deviation would make the ablation meaningless.

3 Implementation

The whole system is written in PyTorch, with the environment remaining in JAX. The world model itself is one self contained file of roughly eight hundred lines. The remaining files are experiment runners, the downstream agent, and tests.

3.1 Architecture

Encoder. Three convolutional layers in the standard Nature DQN configuration, followed by a linear projection to 256 dimensions and a batch normalisation layer. The encoder sees a single RGB frame resized to 84×84 .

Predictor. A four layer causal transformer over the sequence of frame embeddings. Actions enter through adaptive layer normalisation with a zero initialised modulation head, as in the paper. Each action produces per token shift, scale and gate parameters for both sub layers of the block. Because the modulation head is initialised to zero, every gate starts at zero and the entire predictor is exactly the identity function at initialisation. The predictor therefore begins as a no operation and learns to use actions, rather than injecting randomly scaled action noise into the residual stream from the first step. The output passes through a projector matching the encoder’s, a linear layer followed by batch normalisation.

Loss. Mean squared error between predicted and target embeddings, plus $\lambda = 0.1$ times the SIGReg term, with gradients flowing through the targets.

3.2 Deviations from the paper

Three deviations were made, all deliberate and all argued here rather than hidden.

First, the encoder is a convolutional network rather than a vision transformer. Atari frames are small and visually simple, a convolutional trunk is considerably cheaper on the hardware available, and the paper itself reports that the method is encoder agnostic. This is the deviation most likely to matter and it is the one that would be reverted first if more compute were available.

Second, data is collected online from a uniform random policy rather than from a prepared offline reward free dataset. The lab provides environments, not datasets, so an online collector was the natural choice. The consequence is that the model only ever observes the part of the state space a random agent reaches, which is a severe restriction on games with sparse rewards such as Montezuma’s Revenge.

Third, and most significantly, the trained predictor is never used for control. The paper evaluates LeWM through model predictive control in the latent space. In this work the predictor is trained and evaluated as a predictor, and the downstream agent learns a policy on top of the encoder instead. Latent planning was not implemented and is the clearest piece of unfinished work.

3.3 Data collection

Episodes are played under a uniform random policy and cut into non overlapping windows of seventeen frames, sixteen for context and one as the final target. Windows never straddle an episode boundary. Episodes run for up to five hundred environment steps and the windows kept from a long episode are sampled uniformly across it rather than taken from the front, so that coverage is not restricted to the opening seconds of a game. Section 9 explains why this matters and how the earlier version of the collector was wrong.

4 Experimental setup

All experiments in this report were run on a single laptop. The relevant details are given in Table 1, since the lab guidelines ask for information about the training system.

Table 1: Hardware and software used for all reported results.

Machine	Apple M5, 32 GB unified memory
Python	3.11.16
PyTorch	2.12.0, running on the Metal (MPS) backend
JAX	0.10.1, running on CPU

One property of this setup shaped the entire software design and is worth stating explicitly. JAX has no GPU backend on Apple silicon and therefore runs on CPU, while PyTorch reaches the GPU through Metal. A benchmark of the Nature CNN trunk at batch size 512 measured one forward and backward pass at 772 milliseconds under JAX on CPU against 37 milliseconds under PyTorch on Metal, a factor of roughly twenty on the operation that dominates policy optimisation. The agent was originally written in JAX and ran at eight environment steps per second. Rewriting the networks in PyTorch while leaving the environment in JAX raised this to roughly 1300 steps per second. This is a property of the hardware rather than of either framework, and on a CUDA machine the natural choice would be different.

World model training uses 5000 gradient steps per game, a replay buffer that grows from 600 to 2600 sequences, and the Adam optimiser at a learning rate of 3×10^{-4} . Agents are trained for one million environment steps with hyperparameters matching the repository’s own PPO configuration, so that the comparison against the repository baseline is meaningful.

5 Evaluating a world model that predicts embeddings

Before presenting results, the evaluation protocol needs justification, because the obvious metric is actively misleading here.

The training loss cannot be trusted. A collapsed encoder achieves a *lower* prediction loss than a healthy one, because predicting a constant is trivial. Any evaluation that reads the loss curve and concludes that training went well will therefore rate a completely useless model as the best one. This is not a hypothetical concern. It is exactly what the ablation in Section 7 demonstrates.

Two diagnostics are used instead.

Effective rank. The entropy based effective rank of the matrix of embeddings, out of a maximum of 256. This is the correct collapse metric for this architecture specifically because the encoder ends in batch normalisation, which pins the variance of every individual dimension to one. Collapse therefore cannot show up as shrinking embedding norms, which is the naive thing to measure. It shows up as the embeddings occupying a low dimensional subspace, and that is what effective rank detects. A value near one means total collapse.

Open loop rollout error against a trivial baseline. The predictor is given three true embeddings and then run autoregressively on its own outputs for fourteen further steps. The resulting mean squared error is compared against a trivial predictor that simply assumes the embedding never changes. Because mean squared error in a learned latent space has no absolute scale, and because different training configurations produce differently scaled latent spaces, only the ratio between the model and its own baseline is comparable across configurations. All rollout numbers in this report are therefore reported as that ratio, where a value of one means no better than assuming nothing happens.

6 World model results

The model trains stably on all fifteen games. Effective rank settles between 22 and 47 out of 256 on every game, far from the value near one that indicates collapse, so the anti collapse mechanism does its job everywhere in the suite.

Fourteen of the fifteen games beat the trivial baseline. Ms. Pacman does not, at a ratio of 1.037, and Gravitar only barely does at 0.946. Ms. Pacman is reported here as a failure rather than quietly dropped from the table. The likely explanation is that both games have visually busy screens with many independently moving sprites under a random policy, which makes the next embedding genuinely hard to predict, while the trivial baseline is correspondingly strong because most of the screen does stay the same between frames. This is a plausible reading rather than a tested one.

It is worth being careful about the phrase "no collapse". Kangaroo at effective rank 25 is using about ten percent of the available embedding space. The defensible claim is that the rank stays far above one, and specifically far above the value of 1.7 reached when the regulariser is removed, not that the representation is using the space generously.

Table 2: World model after 5000 training steps on each of the fifteen required games. The ratio column is open loop rollout error divided by that model’s own frozen baseline, so lower is better and 1.0 means no better than assuming the embedding never changes.

Game	Effective rank	Rollout ratio	Beats baseline
Asteroids	32.9	0.550	yes
BeamRider	33.1	0.227	yes
Breakout	27.6	0.164	yes
Enduro	22.1	0.368	yes
Freeway	47.0	0.058	yes
Frostbite	26.5	0.460	yes
Gravitar	22.9	0.946	yes
Kangaroo	25.0	0.315	yes
Montezuma’s Revenge	29.0	0.364	yes
Ms. Pacman	28.3	1.037	no
Phoenix	28.9	0.877	yes
Pong	28.4	0.273	yes
Seaquest	28.5	0.236	yes
Skiing	40.2	0.275	yes
Tennis	27.7	0.509	yes

7 Ablations

The ablations are the most informative part of this work. Each variant changes exactly one thing relative to the faithful configuration and is trained identically otherwise, on Pong and on Seaquest.

Table 3: Ablations after 5000 steps. Values are given as Pong / Seaquest. The rollout column is the ratio to each variant’s own frozen baseline, for the reason given in Section 5.

Variant	Effective rank	Prediction loss	Rollout ratio
Faithful (the paper)	28.4 / 28.5	0.0087 / 0.0113	0.286 / 0.235
No SIGReg ($\lambda = 0$)	1.7 / 1.9	0.0010 / 0.0004	1.006 / 0.695
Stop gradient plus SIGReg	174.4 / 68.6	0.0730 / 0.0526	0.379 / 0.356
Stop gradient only	149.4 / 42.8	0.0960 / 0.0942	0.430 / 0.689
Additive action conditioning	21.2 / 21.1	0.0092 / 0.0115	0.310 / 0.188
$\lambda = 0.5$	58.9 / 44.6	0.0209 / 0.0225	0.289 / 0.305
$\lambda = 2.0$	89.9 / 59.5	0.0399 / 0.0393	0.330 / 0.409

7.1 Removing SIGReg collapses the representation

With $\lambda = 0$ the effective rank falls to 1.7 on Pong and 1.9 on Seaquest, which is total collapse. At the same time the prediction loss *improves* by a factor of nine on Pong and twenty eight on Seaquest, reaching 0.0010 and 0.0004 against 0.0087 and 0.0113 for the faithful configuration.

This is the single most useful result in the report, and not because it is surprising. It is useful because it makes the failure mode concrete and quantitative. A practitioner watching only the training loss would conclude that removing the regulariser had improved the model by an order of magnitude. The downstream consequence is that on Pong the collapsed model then fails to beat even its own trivial baseline, at a ratio of 1.006. The model has learned nothing and the loss curve says it has learned a great deal.

7.2 The paper’s claim, and where it has to be narrowed

The interesting control is the one with $\lambda = 0$ and a stop gradient enabled, which is the BYOL style configuration the paper positions itself against. If that configuration also collapsed, the claim "SIGReg is what prevents collapse" would be straightforwardly supported.

It does not collapse. Effective rank reaches 149.4 on Pong and 42.8 on Seaquest, in both cases far from collapse. A plain stop gradient prevents collapse perfectly well in this setting, without any regulariser at all.

The claim therefore has to be narrowed, and the narrowed version is the one this report makes. SIGReg is *sufficient* to prevent collapse without a stop gradient, which is a real and useful property, but it is not *necessary* for that purpose. What SIGReg contributes over a stop gradient is prediction quality: its prediction loss is roughly eleven times lower on Pong and eight times lower on Seaquest than the stop gradient only configuration, and its rollout ratio is better than either stop gradient variant. Adding a stop gradient on top of SIGReg makes the model worse rather than better.

A related point is worth stating because it is easy to get backwards. High effective rank is not itself a goal. The stop gradient variant has the highest rank of any configuration tested, at 174.4, and a clearly worse rollout ratio than the faithful configuration. Rank is a diagnostic for collapse, not a measure of quality.

7.3 Regularisation strength

Raising λ from 0.1 to 0.5 to 2.0 increases effective rank monotonically and degrades the rollout ratio monotonically on both games, from 0.286 to 0.289 to 0.330 on Pong and from 0.235 to 0.305 to 0.409 on Seaquest. The value of 0.1 recommended by the paper is therefore also the right value on Atari, which is a small piece of evidence that the hyperparameter transfers across domains.

7.4 Action conditioning

Replacing the adaptive layer normalisation conditioning with a simple additive action embedding produces essentially no difference. The adaptive scheme gives a slightly lower prediction loss on both games, while the additive scheme gives a better rollout ratio on Seaquest and a worse one on Pong.

This result is reported rather than omitted, even though it is unflattering to the architecture that was implemented faithfully. The honest summary is that the paper’s conditioning mechanism shows no measurable benefit at this scale, and it is retained in the implementation for fidelity rather than because the evidence supports it. An earlier version of these experiments had a data bug that corrupted a quarter of all action labels, described in Section 9, which would have been a perfectly good excuse for finding no difference. The numbers above come from the corrected pipeline and the conclusion did not change.

8 Downstream evaluation with a reinforcement learning agent

Because LeWM is trained without rewards, it produces no game scores. To obtain numbers comparable with published reinforcement learning results, a PPO agent was trained on top of the learned representation.

8.1 Setup

Three configurations share identical policy and value heads, identical hyperparameters, identical observations and an identical step budget. They differ in where the features come from:

- **scratch**: a Nature CNN trained end to end with the policy. This is the control.
- **frozen**: the pretrained LeWM encoder with its weights frozen.
- **fine tuned**: the pretrained LeWM encoder, updated by the policy gradient.

The world model encodes single frames, so the four frame stack is encoded frame by frame with the same encoder and the resulting embeddings are concatenated, which is what gives the policy access to motion.

One caveat about this comparison should be stated plainly, because the natural way to describe it is too strong. It is not true that the only difference is where the features come from. The scratch network mixes the four stacked frames convolutionally and therefore builds spatially local motion features, whereas the LeWM trunk encodes each frame independently and combines them with a single linear layer. The two trunks have different access to precisely localised motion, and that difference is a plausible contributor to the results on Breakout in particular. It is not controlled for here.

8.2 Results

Table 4: Episodic return after one million environment steps, averaged over the final ten percent of training. Scores are unclipped game scores. Single seed.

Game	PPO from scratch	On frozen LeWM	On fine tuned LeWM
Pong	15.3	17.2	−15.6
Seaquest	949	571	411
Breakout	40.8	12.2	12.0

Taken at face value this table suggests that the pretrained representation helps on Pong and hurts elsewhere. The next section shows that the first half of that reading does not survive contact with multiple seeds.

8.3 Three seeds on Pong

A single reinforcement learning run on Atari carries very little information, so Pong was repeated with three seeds.

Table 5: Pong over three seeds. The two columns measure different things and are deliberately kept apart.

	Final return	Steps to first positive return	
PPO from scratch	12.60 ± 7.27	0.61M	[0.63, 0.69, 0.51]
PPO on frozen LeWM	17.36 ± 1.09	0.32M	[0.27, 0.40, 0.29]

The final score difference is not significant. The gap between the means is 4.8 while the standard deviation of the scratch configuration alone is 7.3. A claim that LeWM produces higher final scores on Pong does not survive three seeds, and no such claim is made in this report. The single seed table above would have supported exactly that claim, which is a good illustration of why the extra runs were worth the compute.

Sample efficiency is a clean result. The number of steps required to reach positive play is roughly halved, and the per seed values do not overlap at all. The worst LeWM seed, at 0.40M steps, still reaches positive play sooner than the best scratch seed at 0.51M. Three out of three. This is the effect that a pretrained representation is supposed to produce, and unlike the final score it holds up under repetition.

Variance falls sharply. The standard deviation drops from 7.27 to 1.09, roughly a factor of seven. Every LeWM seed lands within a two point band while the scratch runs span fourteen points. The pretrained representation makes the outcome of a run predictable as well as faster, which for a practitioner is worth something on its own.

8.4 Where the representation does not help

On Seaquest and Breakout the frozen encoder plateaus clearly below the scratch network. The obvious explanation is that the encoder was trained on data from a random policy and therefore goes out of distribution as the policy improves. That explanation is testable: unfreezing the encoder should allow it to adapt and close the gap.

It does not. The fine tuned configuration improves Seaquest and Breakout only marginally, and on Pong it is dramatically worse than leaving the encoder alone. The distribution shift explanation is therefore not supported by the experiment designed to test it, and this report does not advance it.

An alternative explanation, which is stated here as a hypothesis and not as a finding because it has not been tested, is an information bottleneck. LeWM is optimised to predict, and prediction does not require the information that control requires. The arrangement of remaining bricks in Breakout barely affects the next frame but entirely determines the achievable score. Pong may be the exception precisely because its state essentially is the ball and two paddles, so almost everything prediction preserves is also what control needs. A linear probe, measuring how well task relevant quantities can be decoded from the frozen embedding compared with features from the scratch network, would test this directly and is the most obvious next experiment.

One further caveat on the fine tuned arm. Pretrained encoder weights are updated at one tenth of the learning rate used for the freshly initialised heads. At the full rate training diverges, for reasons explained in Section 9, while at the reduced rate the encoder adapts only slightly. Fine tuning here is caught between instability and inertia, so the result should be read as "does not close the gap at this setting" rather than as a general statement about fine tuning.

8.5 Comparison against published scores

The lab guidelines ask for a comparison against reported PPO scores from the original ALE implementation. The source they link is the original PPO paper of Schulman et al. (2017), whose Table 6 reports mean final scores over the last one hundred episodes after forty million game frames, equivalently ten million timesteps at a frame skip of four.

The budget difference has to be read alongside these numbers. The agents here are trained for one million environment steps, which is one tenth of the budget behind the published column, and published Atari results are elsewhere often reported after two hundred million frames, which is a factor of fifty. Nothing in this table should be presented as a converged comparison.

With that qualification stated, the pattern is still informative. On Pong and Seaquest this implementation reaches roughly four fifths of the published score using one tenth of the interaction, which is the behaviour expected of a correct implementation stopped early. Breakout is the clear outlier at fifteen percent, and it is the same game on which the frozen representation performs worst. Breakout rewards precise tracking of a small fast object, and the limitation noted in Section 10 applies with particular force there: the LeWM trunk encodes each frame independently and concatenates the embeddings, so it has weaker access to precisely localised

Table 6: This work against the published PPO scores of Schulman et al. (2017), Table 6. The published column is measured at ten times the step budget used here, so the columns are not directly comparable and the ratio in the final column is given to make the size of that gap explicit rather than to hide it.

Game	Scratch (1M)	Frozen LeWM (1M)	Published PPO (10M)	Ratio (best here / published)
Pong	15.3	17.2	20.7	0.83
Seaquest	949	571	1204.5	0.79
Breakout	40.8	12.2	274.8	0.15

motion than a network that convolves over the stacked frames. That this is also the game where the gap to published PPO is widest for *both* arms suggests the budget is the dominant factor and the representation a secondary one, but the two are not separated here.

9 Methodological problems found during the work

Three problems were found and fixed during the project. None of them caused a crash. All three produced plausible looking numbers that were wrong, which is the category of bug that is most dangerous in empirical work, so they are documented here rather than quietly repaired.

9.1 Corrupted action labels from sticky actions

The environment wrapper used for world model data collection took its default setting of a 0.25 sticky action probability, while the wrapper used for the agent explicitly disabled it. Under stickiness the environment executes the *previous* action with probability one quarter, but the calling code only ever sees the action it requested. The world model was therefore learning $p(z' | z, a)$ from action labels that were wrong a quarter of the time.

This directly corrupted the experiment intended to measure action conditioning, and it also meant the encoder was pretrained under different dynamics than it was later deployed on. Both environments now disable sticky actions.

9.2 Coverage restricted to the opening seconds of each game

Episodes were capped at sixty eight environment steps, and the windows kept from each episode were taken from the front. The model therefore never observed any part of a game beyond roughly the first four and a half seconds of play.

Investigating this revealed a second cause that was easy to miss. On games with short lives the cap was not even the binding constraint. With episodic life handling enabled, an episode ends at the first lost life, which under a random policy is about twenty two steps in Breakout. The model could not have seen a partly cleared brick wall regardless of the cap.

Both were fixed. Episodic life handling is disabled for reward free collection, where it serves no purpose, episodes now run up to five hundred steps, and windows are sampled across the whole episode. Measured effect: Breakout episodes went from twenty two steps to two hundred and seventeen, Pong and Seaquest from sixty eight to five hundred and four hundred and thirty four.

This matters for interpretation and not only for tidiness. A model that has never seen a partly cleared brick wall has a coverage based explanation for its poor Breakout performance, and that explanation was confounded with the information bottleneck hypothesis discussed above.

9.3 Batch normalisation inside a PPO policy trunk

When the encoder was fine tuned it remained in training mode, so its batch normalisation layers normalised using batch statistics during the policy update but running statistics during the rollout, with batch sizes differing by a factor of thirty two between the two.

This interacts badly with PPO specifically. PPO stores action log probabilities during the rollout and recomputes them during the update in order to form an importance ratio. If the two passes are computed by functions that differ, the ratio is meaningless. Training diverged, with the value loss reaching 10^{15} and the policy entropy collapsing to zero within twenty five iterations on every game.

The fix is to keep the encoder’s normalisation layers in inference mode at all times, so that only the weights are updated. This removed the divergence but introduced the second problem noted earlier: with the normalisation statistics fixed, nothing constrains the scale of the encoder’s output, and at the full learning rate the embeddings drift until the policy logits saturate. Hence the reduced learning rate for pretrained weights. The general lesson, which generalises beyond this project, is that batch normalisation inside a policy trunk is specifically hazardous for algorithms that compare two forward passes.

10 Limitations

- **Compute budget.** Agents are trained for one million environment steps against a reference of two hundred million. The curves show learning trends and not converged performance.
- **Seeds.** Pong uses three seeds. Every other agent number in this report comes from a single seed and should be read as indicative only. The Pong result demonstrates why this matters, since one claim survived the addition of seeds and another did not.
- **The agent comparison is not purely a comparison of features**, for the architectural reason given above.
- **Random collection policy.** The world model observes only the states a random agent reaches, which is a small part of the state space on sparse reward games.
- **Evaluation detail.** The rollout evaluation keeps batch normalisation in batch statistics mode for consistency with training. Using fixed running statistics instead changes the rollout error by about seven percent, measured on Pong. That figure is the size of the methodological uncertainty underneath the rollout numbers in this report.
- **Ms. Pacman fails** the rollout evaluation and this is not explained.
- **The predictor is never used for control**, which is the paper’s own evaluation method.

11 Conclusion

LeWorldModel was implemented from scratch for JAXAtari and trains stably across the full fifteen game suite without representation collapse, producing useful open loop latent predictions on fourteen of the fifteen games.

The ablations support a narrowed version of the paper’s central claim. Removing SIGReg collapses the representation completely while improving the training loss by up to a factor of twenty eight, which makes the point that this objective cannot be monitored through its loss. However, a plain stop gradient also prevents collapse, so SIGReg is sufficient rather than necessary for that purpose. Its actual contribution is prediction quality, where it beats the stop gradient configurations by roughly an order of magnitude.

Downstream, the pretrained representation roughly halves the environment steps needed to reach positive play on Pong, with no overlap across three seeds, and reduces run to run variance by a factor of about seven. It does not produce a significantly higher final score, and on Seaquest and Breakout it plateaus below an equivalent agent trained from pixels. Fine tuning does not close that gap, which rules out the most obvious explanation and leaves an open question.

The most valuable extensions, in order, would be latent model predictive control so that the predictor is evaluated the way the paper intends, a linear probe to test the information bottleneck hypothesis, and additional seeds on the remaining games.

Code and reproducibility

The implementation lives in `scripts/benchmarks/jepa/` in the JAXAtari repository. The world model is a single self contained file. Eighteen unit tests cover the properties that fail silently rather than loudly, including the collapse diagnostic, predictor causality, and two regression tests for bugs described in Section 9. Every experiment in this report can be reproduced with the commands documented in the accompanying README, with a fixed seed.

Use of AI assistance

Claude (Anthropic) was used throughout the implementation and analysis, in line with the course’s stated position on the use of language models. It contributed to writing and debugging code, to designing the evaluation protocol, and to drafting this report. All experiments were run and inspected by the author, and the numbers reported here come from the recorded output of those runs. Three of the corrections described in Section 9 were identified during review of the generated code rather than at the time of writing, which is itself a reason to treat generated experimental code with the same suspicion as any other.

References

1. Maes, Le Lidec, Scieur, LeCun and Balestriero. *LeWorldModel: Stable End to End Joint Embedding Predictive Architecture from Pixels*. arXiv:2603.19312, 2026. Reference implementation: <https://github.com/lucas-maes/le-wm>
2. Schulman, Wolski, Dhariwal, Radford and Klimov. *Proximal Policy Optimization Algorithms*. arXiv:1707.06347, 2017.
3. Huang et al. *CleanRL: High quality single file implementations of deep reinforcement learning algorithms*. The PPO implementation here follows the repository’s adaptation of CleanRL.
4. Delfosse et al. JAXAtari. <https://github.com/k4ntz/JAXAtari>

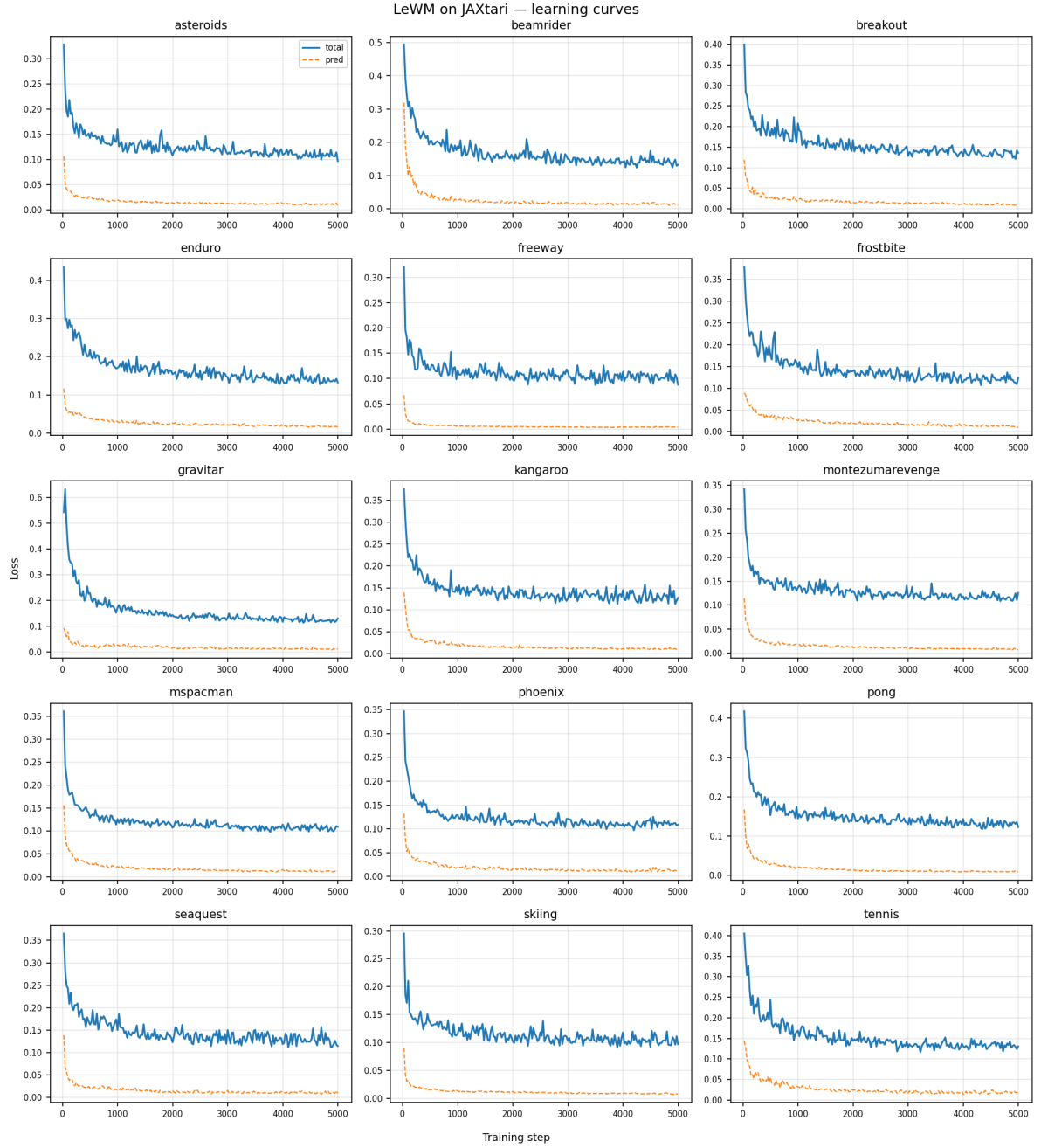


Figure 1: Training curves for all fifteen games. Loss falls smoothly everywhere, which as argued in Section 5 is not by itself evidence that anything useful was learned.

LeWM ablations — does SIGReg alone prevent collapse?

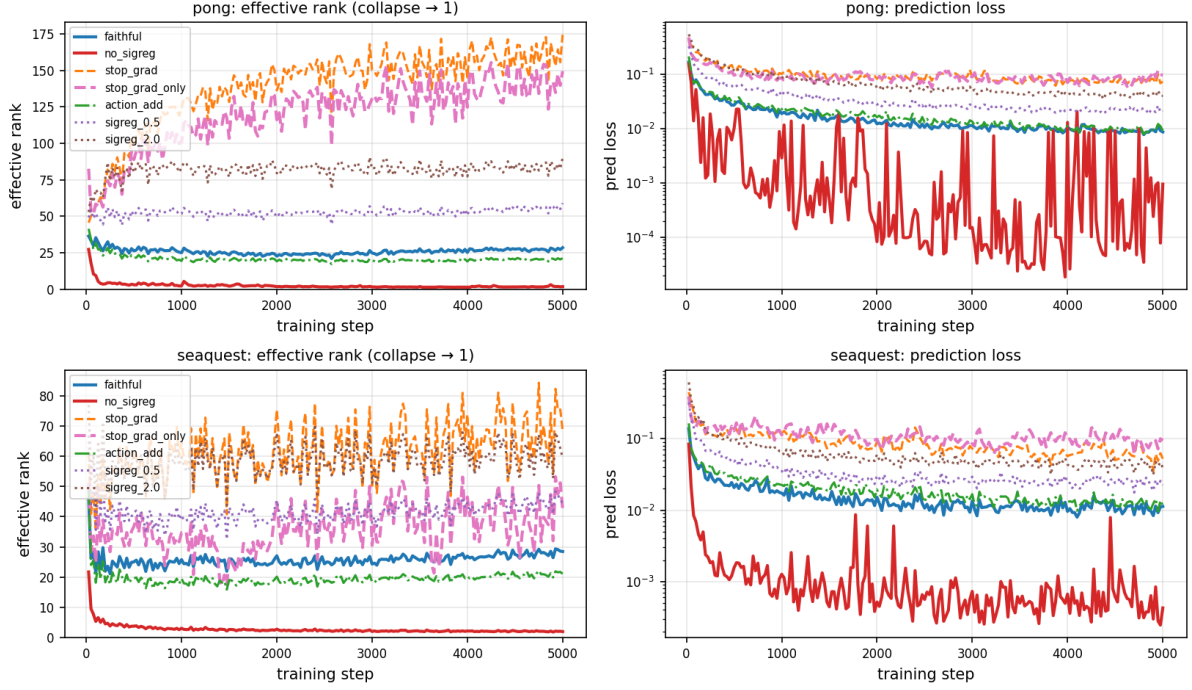


Figure 2: Effective rank and prediction loss over training for every ablation. The red curve is the configuration without SIGReg. Note that it has both the lowest rank and the lowest loss, which is the entire point.

Does a LeWM representation help PPO? (identical budget per arm)

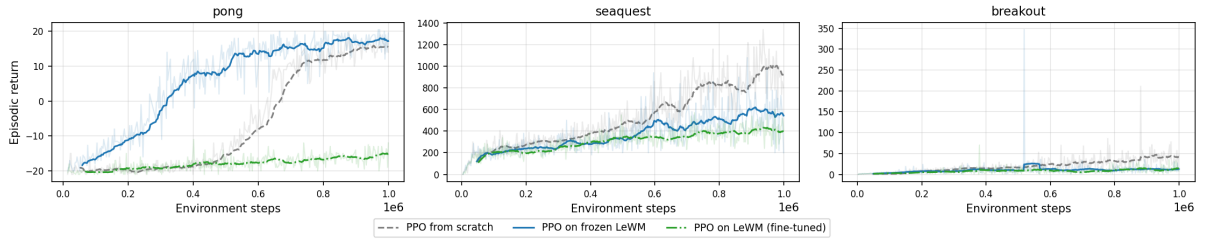


Figure 3: Learning curves for the three configurations on each game, single seed. The pretrained representation leads early on Pong, tracks the baseline on Seaquest before falling behind, and plateaus well below it on Breakout.

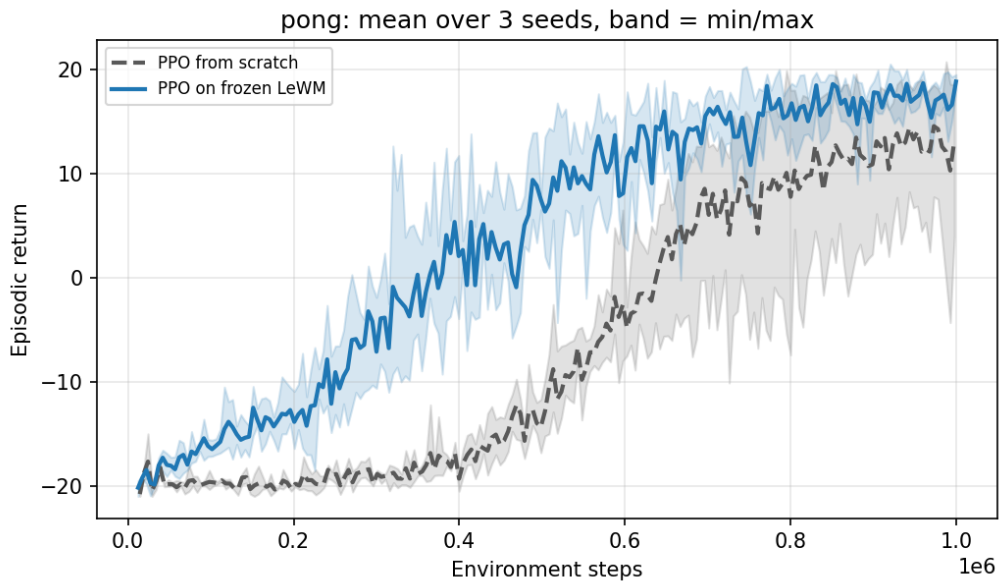


Figure 4: Pong over three seeds. The line is the mean and the band spans the minimum and maximum over seeds. The two configurations separate early and converge late, which is the shape a sample efficiency gain has.